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Roll No

IT-710-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Advanced Concepts in Database Systems

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. What is the meant by Query processing? Discuss the various methods for query optimization with the help of suitable example.
2.
 - a) Draw an ER diagram for library management system.
 - b) Discuss data transfer costs of distributed query processing.
3.
 - a) Explain about Complexity of relational algebra operations.
 - b) Explain briefly about Fragmentation with suitable examples.
4. What is the purpose of Indexing? Discuss about:
 - a) Primary and secondary indices
 - b) Dense and sparse indices
5.
 - a) How timestamp based concurrency control is different from locking schemes? Explain.
 - b) What is concurrency control? How is it implemented in DDBMS? Explain.

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6.
 - a) Explain about Temporal database.
 - b) Explain about deductive database.

7.
 - a) Differentiate between spatial and multimedia database with suitable example.
 - b) Discuss the issues related to real time database design.

8. Write short notes on:
 - i) XQuery
 - ii) JDBC
